

Reducing false positives in fraud detection: Combining the red flag approach with process mining

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ABSTRACT

Fraud detection often includes analyzing large datasets of enterprise resource planning systems to locate irregularities. Analysis of the datasets often results in a large number of false positives, that is, entries wrongly identified as fraud. The aim of our research is to reduce the number of false positives by combining the red flag-based approach with process mining. The red flag approach presents hints for unusual behavior, whereas process mining reconstructs and visualizes the as-is business process from the underlying dataset. The combination of these two techniques allows for identification and subsequent visualization of possible fraudulent process instances with the corresponding red flags. We exemplarily applied our new approach to the purchase-to-pay business process to successfully identify 15 of 31 fraud cases in our dataset. Our false positive rate was 0.37%, which is considerably lower than rates reported in similar research papers.

1. Introduction

Recent surveys regarding fraud show that fraudulent behavior is a worldwide issue, consuming an estimated 5% of the typical organization's annual revenue (ACFE, 2016). The evolution of computer-assisted audit tools and techniques enables auditors to retrieve and analyze large amounts of data from enterprise resource planning (ERP) and linked systems (Bönner et al., 2011; Coderre, 2009). These tools paved the way for a variety of data mining-based techniques for detecting fraud such as neural networks, Bayesian networks, decision trees, regression models and genetic algorithms (Bolton and Hand, 2002; Ngai et al., 2011; Phua et al., 2010). These data mining approaches are inductive fraud detection approaches. Inductive fraud detection often results in a flood of alarms including many false positives: that is, entries that look unusual but are not fraud (Albrecht et al., 2012; Alles et al., 2006). Analyzing false positives is time-consuming and consequently causes costs without realization of benefits (Luell, 2010).

A further issue of the proposed data mining approaches is that they do not utilize sequential information (Phua et al., 2010). Since currently established ERP systems are inherently process-oriented (Aalst, 2005; Boczeko, 2007), ignoring sequential information for algorithmic fraud detection will restrict and significantly limit the use of inductive fraud detection approaches. Process mining directly builds upon sequential and temporal dependencies within process instances (Aalst et al., 2005). It recreates and visualizes the as-is business process by extracting information from the underlying dataset (Aalst, 2011).

We aim to combine a deductive approach with process mining to counteract the deficits inherent to inductive fraud detection approaches. A typical deductive approach for fraud detection is the red flag approach, recommended by most fraud auditing standards (e.g. the Statements of Auditing Standards [SAS]), which provide guidance to auditors on auditing a company (Albrecht et al., 2012). A red flag is a set of circumstances that are unusual in nature or deviate from the normal activity signifying that something is out of the ordinary and may need further investigation (DiNapoli, 2008). We do not restrict our search to finding incoherent red flags,

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but rather simultaneously scan our dataset for a combination of red flags, which we call “fraud patterns”. We assume that searching for red flags belonging to a particular fraud case can reduce the number of false positives. The idea of the fraud pattern is taken from [Gamma et al. \(1995\)](#), who describe software design patterns. These patterns represent a reusable solution to a common problem within a given context in software design. Analogue fraud detection patterns represent a reusable solution to identify a specific fraud case in a dataset of a business process.

In our research, we combine the red flag approach with process mining. Our guiding research question is the following:

Can a combination of red flags and process mining reduce the number of false positives in fraud detection?

To evaluate our approach, we developed a prototypical implementation and applied it to a purchase-to-pay business process.

To the best of our knowledge, no research has been conducted to date combining the red flag-based approach with process mining to visualize fraudulent process instances with the corresponding red flags to reduce the amount of false positives.

2. Theoretical background

In this section, we provide an overview of current literature defining fraud and the corresponding Association of Certified Fraud Examiners (ACFE) tree for classification of fraudulent activities. We introduce process mining and the red flag approach for fraud detection and conclude with the description of the purchase-to-pay business process as applied in our prototype.

2.1. Fraud

Scholars and practitioners have used and still use numerous terms to describe different forms of fraudulent behavior. We concentrate on occupational fraud defined as a “crime committed in the course of [the] occupation.” ([Clinard and Quinney, 1967](#)) and use the [ACFE \(2016\)](#) definition of occupational fraud: “the use of one's occupation for personal enrichment through the deliberate misuse or misapplication of the employing organization's resources or assets” ([ACFE, 2016](#)).

One de-facto standard for structuring and classifying fraud cases is the fraud tree introduced by [Wells \(2011a, 2011b\)](#) and the [ACFE \(2016\)](#), which is the worldwide largest organization of fraud examiners and publishes an annual “Report to the Nations” presenting current facts and figures on fraud. An excerpt of the fraud tree, including fraud cases that can occur within the purchase-to-pay business process, is shown in [Fig. 1](#).

Corruption, asset misappropriation and fraudulent statements are at the highest level of the fraud tree ([Wells, 2011a, 2011b](#)). Both corruption and asset misappropriation are transactional in nature and include the theft or intentional misuse of assets or the abuse of one's position. Fraudulent statements are defined as a deliberate misrepresentation of a company's financial statements, timing differences or improper disclosures.

Conflict of interests, bribery, illegal gratuities and economic extortion are included within the category of corruption. A conflict of interest exists when an executive or an employee has undisclosed economic or personal interest in a business transaction and influences a transaction to the disfavor of the company ([Wells, 2011a, 2011b](#)). Typical examples are purchasing schemes in which a department enters and maintains business relationships with third parties potentially resulting in a conflict of interest ([Wells, 2011a, 2011b](#)). While bribery involves accepting, giving or even offering anything of value to influence a transaction ([ACFE, 2016](#)), illegal gratuities change hands ex-post and are considered as a reward and not as exertion of influence. Economic extortion occurs when employees or managers actively demand something of value in exchange for a favorable decision. The person offering a bribe essentially buys influence in decision-making ([Wells, 2011a, 2011b](#)). There are two common types of bribery schemes: kickbacks and bid rigging ([ACFE, 2016](#)). In the so-called kickback scheme, a percentage of income is paid to a person in a position of power or influence as payment for having made the income possible. While the vendor issues and submits invoices that are usually either inflated or totally fictitious, the employee makes sure the invoices are approved for payment and a share of the generated surplus is diverted back ([Wells, 2011a, 2011b](#)). Bid rigging occurs when vendors are willing to pay for influencing a competitive bidding process ([Wells, 2011a, 2011b](#)) and ranges from swaying the specification in the pre-solicitation phase to making competitors' bids accessible before the submission phase ends ([ACFE, 2016](#)).

Within asset misappropriation, a differentiation is typically made between the misappropriation of cash and the misappropriation of inventory and other assets ([Wells, 2011a, 2011b](#)). Typical examples of asset misappropriations include skimming revenues,

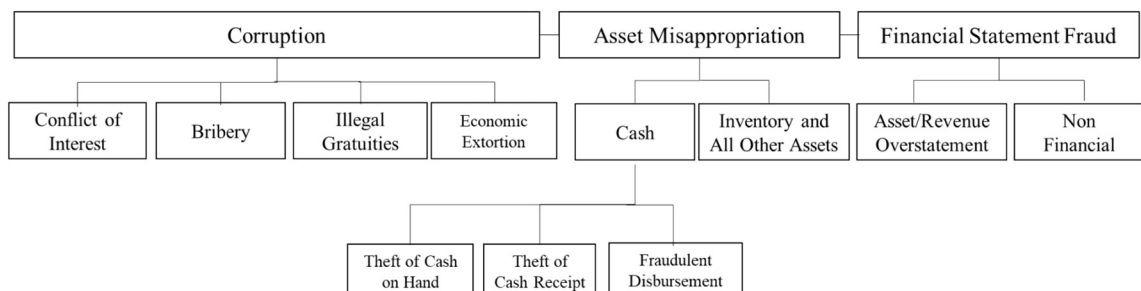


Fig. 1. Classification of occupational fraud and abuse for fraud in the purchase-to-pay business process, based on [ACFE \(2016\)](#) abridged version.

stealing inventory and payroll fraud. In cash misappropriation, billing schemes are targeted activities within the purchase-to-pay process. There are three different billing schemes: shell companies, non-accomplice vendors and personal purchases. Shell companies are entities that do not engage in conventional economic activities but serve the sole purpose of establishing a scheme to commit fraud (Wells, 2011a, 2011b). Shell companies submit inflated invoices, deliver goods of lower quality than ordered, or invoice fictitious orders (ACFE, 2016).

Although less frequently observed, another fraud is to fool legitimate vendors to unknowingly take part in an illegitimate billing scheme (Wells, 2011a, 2011b). Since the perpetrator does not have access to payments received by suppliers, the so-called non-accomplice vendor schemes take advantage of refunds issued by the vendors (ACFE, 2016). In order to trigger suppliers to send refunds, the fraudster can simply pay vendors twice or redirect payments to the wrong supplier, negotiate refunds for the 'mistakenly' paid amounts and pocket those (Wells, 2011a, 2011b). The fraudster could also tamper with the ordered quantity of goods and return the excess goods while pocketing the refund, or simply overpay the supplier waiting for the excess amount to be refunded (ACFE, 2016).

Personal purchases constitute the third type of billing scheme. They cover purchases that follow the standard approval process including purchase requisitions and purchase orders, as well as purchases against charge accounts (Wells, 2011a, 2011b). Ensuring the approval of purchase requisitions, purchase orders and payments is essential to the success of these schemes (Wells, 2011a, 2011b). Moreover, perpetrators can use corporate credit cards to buy personal goods or return legitimately purchased goods for cash (ACFE, 2016).

2.2. Red flags and process mining

The red flag-based approach is a well-established technique in fraud detection and is recommended by most auditing standards (Albrecht et al., 2012). Red flags are hints or indicators for fraudulent behavior and show that something irregular has happened. A red flag is not proof of fraud as there might be a sound explanation for the existence of the indicators (Albrecht et al., 2012).

Fraud should be distinguished from anomalies. While anomalies occur evenly distributed over all process steps, fraud is conducted intentionally. The perpetrator tries to cover up his tracks, so red flags of fraud may be found in very few transactions – an act of searching for the proverbial needle in the haystack (Albrecht et al., 2012).

Every user action leaves traces in the system. These audit trails are generally automatically stored in the system. Datasets are then analyzed using structured query language (SQL) to identify process instances where these red flags occur (Coenen, 2008; Stamler et al., 2014). Sources analyzed include ERP systems, document management or supply chain systems. In addition, paper-based sources like original receipts may be taken into account (Albrecht et al., 2012). For a more extensive data analysis, telephone or e-mail conversations can be analyzed (Islam et al., 2010).

Process mining entails discovering, monitoring, and improving real processes through the extraction of information from event logs of IT systems (Aalst, 2011). Process mining has been previously applied in fraud detection research. The research in this field may be split into two main streams: process deviation analysis and conformance checking.

Researchers have attempted to identify deviations from standard business processes and various approaches have been developed to recognize such process deviations. Bose and Aalst (2010), for example, developed a framework that compares infrequent process instances and combines similar instances into patterns. This approach was later implemented into a live system of a company. The system raises flags if the process instance deviates from the standard business process (Aalst et al., 2010a; Aalst et al., 2010b). Swinnen et al. (2012) proposed a semi-automatic process deviation method to identify internal control weaknesses using process mining and association rule mining. Association rule mining groups deviating cases into business rules according to similar attribute values. Nguyen et al. (2014) have compared different classification techniques from data mining to analyze deviant business process execution. They applied these techniques to six real-life event logs and were able to show that pattern mining techniques slightly outperform techniques that are based on individual activity frequency.

A second similar research string is conformance checking, which identifies missing segregation of duties or a missing four-eye principle. In conformance checking, roles and users, including their system rights, are analyzed. For example, a purchase order should not be created and released by the same person. Jans et al. (2014) were able to detect multiple violations of conformance checking in a dataset from a European bank; many of these violations were not identified using traditional auditing methods.

The literature provides multiple algorithms for process extraction. Identifying a process model is a non-trivial issue as a dataset can contain erroneous event logs (noise in the data), rare activities, loops and dependencies (Aalst et al., 2004; Gupta, 2014). Most available process mining algorithms can be classified into three categories: heuristic mining algorithms, genetic mining algorithms, and fuzzy mining algorithms (Gupta, 2014). Techniques more recently developed are the inductive miner and alignment-based techniques.

- (1) Heuristic algorithms attempt to reconstruct the most probabilistic graph. For each activity, the algorithm calculates the probability of a second activity being dependent on the previous one (Saravanan and Rama Sree, 2011).
- (2) Genetic mining algorithms are non-deterministic. They create a graphical representation of the underlying process because of the evolutionary improvement (Gupta, 2014). The best model is selected and continuously refined (De Medeiros and Weijters, 2005).
- (3) Fuzzy mining algorithms allow users to adapt the result to their individual demands by being able to zoom-in and -out. These algorithms create easy representations of the data graph (Günther and Aalst, 2007). The fuzzy miner does not structure the data beforehand, but rather uses methods from thematic cartography (Günther, 2009). The main goal is to abstract from details and visualize the most frequent process instances. Seven metrics are used to measure significance and correlation. Significance

Table 1
Standard purchase-to-pay business process.

1. Create purchase requisition	A need for a good or service is identified within a department and a purchase requisition is created. Optionally, an authorized user should release the purchase requisition.
2. Select vendor	The purchasing department selects a supplier. Framework contracts with a supplier may exist. A framework contract includes general agreements governing terms and conditions for making specific purchases (call-offs).
3. Send purchase order	After vendor selection, a purchase order is sent to the supplier. Again, an authorized user may release the purchase order.
4. Receive goods	The supplier delivers the goods, which are then usually recorded in the system with reference to the purchase order.
5. Receive invoice	An invoice is received for the cost of the goods or services. This invoice is recorded in the system with reference to the purchase order and goods receipt. In some ERP systems, the quantity and price of the delivered goods are checked automatically for discrepancies in the so-called three-way match.
6. Pay invoice	The last step is payment of the invoice.

calculates the frequency of event occurrences and their order: the more frequent a precedence relation is observed, the more significant it is. Correlation measures how closely events are related by analyzing the data that events share or the similarity of activity names. The model only displays highly significant behavior if the behavior is highly correlated as well (Jans et al., 2011). Less significant events are aggregated into clusters and are omitted if the behavior is weakly correlated. Based on these measures, views on the event log can be generated dynamically.

- (4) Inductive miner is able to handle life cycle data and distinguish between concurrency and interleaving. The inductive miner uses the divide and conquer approach to recursively split the event log until each sub log contains only a single activity (Leemans et al., 2015).
- (5) Alignment based techniques try to identify for each trace in the log the closest corresponding trace produced by the log model to determine where the model and the traces diverge (Adriansyah et al., 2012).

2.3. Purchase-to-pay business process

We describe the combination of the red flag approach with process mining on the purchase-to-pay business process because this process is comparatively standardized and a part of virtually every business. Purchasing departments enter and maintain business relationships with third parties; while this is a “normal” business situation, it also presents the potential for fraudulent dealings. We briefly describe the purchase-to-pay business process showing the standard process steps in Table 1 (based on Hall, 2011; Porter, 1998).

3. Method

In this section we provide a description of the method we applied in our work. First we describe the overall approach followed and how we derived the fraud detection patterns. Then we describe how we generated semi-realistic fraud data. In a last step, we provide information about our analysis of the dataset and briefly describe our prototype.

3.1. Overall process followed

Albrecht et al. (2012) proposed a method to identify fraud with red flags, which consists of multiple layers: the analytical, technological, and the investigative. The analytical layer involves understanding the business under study, identifying possible frauds that could exist and cataloguing the red flags. The technology layer contains the steps to gather data about the red flags from the underlying dataset (e.g. using SQL) and to analyze the results. The investigative layer involves the investigation of the identified red flags.

We applied the method described by Albrecht et al. (2012) but slightly adapted it in order to include process mining steps as proposed by Bozkaya et al. (2009). The method from Bozkaya et al. (2009) consists of six process steps that describe how to conduct a process mining analysis: (1) log preparation, (2) log inspection, (3) control flow analysis, (4) performance analysis, (5) role analysis and (6) transfer of results.

In the context of applying process mining for fraud detection, the (1) log preparation and (2) log inspection steps are of high importance. Within the log preparation activities, cases and time stamps are identified to recreate the as-is business process. The log inspection steps include gathering statistics about the process and applying a process mining algorithm on the dataset. We integrated these two processes into the method of Albrecht et al. (2012). The method we used is presented in Table 2.

3.2. Identification of fraud detection patterns and red flags

We conducted a literature review based on the guidelines of Webster and Watson (2002) to identify red flags and fraud detection patterns. We used the ProQuest platform to access multiple databases simultaneously. As keywords we used a combination of the search terms “fraud” or “white collar crime” or “misappropriation” or “corruption” or “conflict of interest” or “bribery” or “kickback” or “kick back” or “shell company” in combination with “purchase-to-pay” or “procure-to-pay” or “accounts payable” or “procurement”. We further included keywords that are primarily perpetrated within the purchase-to-pay process and are therefore not

Table 2

Process steps for fraud detection.

Adapted from Albrecht et al. (2012) and Bozkaya et al. (2009).

Analytical steps	(1) Understand the business (process) (2) Identify (theoretically) the existence of possible fraud (3) Catalogue possible red flags (and determine “fraud detection patterns”)
Technological steps	(4) Log preparation (5) Log inspection (6) Use technology to gather data about symptoms (red flags and fraud detection patterns) (7) Analyze results
Investigative steps	(8) Investigate symptoms (9) Follow up

combined with purchase-to-pay terms. These terms were “payment fraud”, “purchasing fraud”, “billing schemes”, “gratuities”, and “extortion”.

To verify the comprehensiveness of our results, all references listed in the identified literature sources were used for backward search. Excluding all articles not addressing fraudulent conduct in the setting of purchase-to-pay, the review process yielded 22 scholarly contributions and 59 articles in the trade literature.

We analyzed the literature to identify red flags and developed fraud detection patterns using the ACFE tree as a basis for structuring our patterns. Most of our identified articles use wording based on the ACFE tree as it is the de-facto standard for categorizing fraud delicts. To ensure the correct coding in our approach, three independent researchers coded the papers and assigned red flags to the ACFE tree structure. After individual coding, we met to discuss differences and reach consensus. In total, we identified eight fraud detection patterns from the ACFE tree as relevant and assigned the corresponding red flags to them. These were: kickback fraud, bid rigging, shell company, double payment, pass through, non-accomplice vendor, redirect payment fraud and private purchases. In total, we identified 142 red flags mentioned in literature, which can be found in [Appendix 1](#).

3.3. Prototype implementation

We developed a prototypical implementation to identify fraud within the purchase-to-pay business process; the development of this prototype is not the focus of this paper. As the prototype should help to identify the fraud detection patterns and visualize red flags on the fraudulent process instance, we briefly summarize its implementation.

We used Celonis Process Mining¹ due to its ability to extend the data structure. This provided us the opportunity to link our red flags table with the process instance and display further information in the GUI of the prototype.

First, we set up the event log structure and created the needed case and activity tables. Activities represent every single process step (e.g. send purchase order, receive invoice) and cases represent the complete process execution. In ERP systems the data belonging to one process is scattered over dozens of tables. Therefore, we created an SQL script to extract the data from the respective database tables of the ERP system and store it into the case and activity tables (Aalst et al., 2010a, 2010b). The SQL script copies the data for all process steps shown in [Table 1](#) as well as respective changes (e.g., change of the purchase order, change of discounts etc.), as changes often signal unusual behavior. In total, we implemented 24 process steps.

Celonis Process Mining uses fuzzy mining to display the discovered model (Kebede, 2015). This allowed us to zoom into the process and identify infrequent process instances and process deviations. The fuzzy miner cannot distinguish between choice and parallelism (Günther, 2009). Rather, it shows all combinations of the process model, although the original data may contain parallelism or choices. The fuzzy miner may further include process flows that do not exist in the event log. As only a few parallel activities take place during purchase-to-pay business processes, these limitations had a minimal effect on our work. The fuzzy miner provides a significant advantage in handling huge data amounts, which is useful as the size stored within an ERP system of a company is typically huge.

Second, we searched for the red flags and fraud detection patterns in our dataset (the tables from the ERP system). We implemented a second SQL script, which screens the dataset for the fraud patterns and red flags described in [Appendix 1](#). All identified red flags were stored in a central table named *Flagged Cases*, as shown in the overview of the architecture of the prototype ([Fig. 2](#)). In addition to single red flags, a reference to the process instance (Case) where the red flag has occurred is stored (CaseId). This allows the linking of cases with red flags and subsequent display of possible fraudulent process instances along with the occurring red flags in the Graphical User Interface (GUI) of the prototype. A reference to the fraud detection pattern from the *Pattern Catalogue* is also stored (PatternId). The *Pattern Catalogue* includes all identified patterns depicted in [Appendix 1](#). The link between the *Pattern Catalogue* and the *Flagged Cases* table allows the filtering of fraud patterns in the GUI. A detailed description of all possible red flags is stored in the *Red Flag Catalogue* table.

We chose this architecture because it is easily extendable. Adding further red flags or fraud patterns does not require a modification of the *Red Flag Catalogue* tables.

Third, we designed a user interface for our prototype (a dashboard). The main advantage of our user interface compared to other

¹ For further information, please refer to <http://www.celonis.com/>.

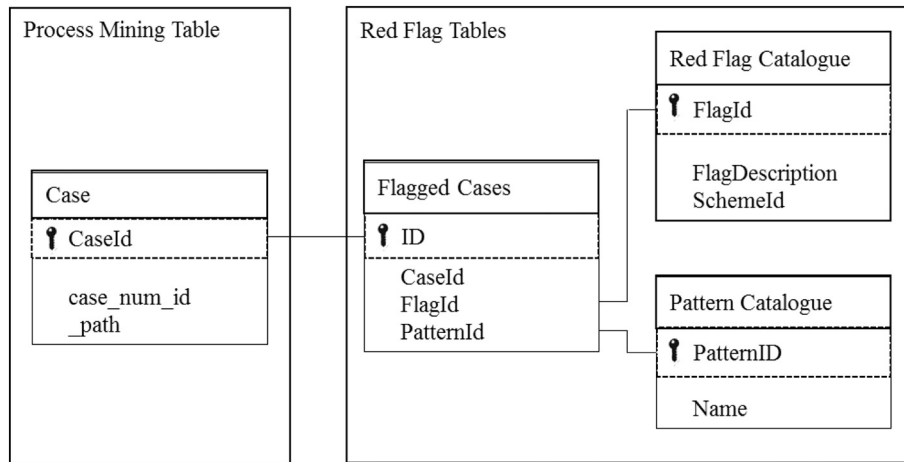


Fig. 2. Architecture of the prototype.

dashboards like Tableau or QlikSense is that it not only includes a dashboard overview of the red flags, but also displays the as-is business process instance along with the occurring red flags and fraud detection patterns. We recommend three applications of the prototype:

- 1) First, the fraud investigator can filter for process instances with the highest number of identified red flags or with the highest possible financial loss. Then the process instances with the most frequent occurring red flags or with the highest financial loss can be analyzed in detail by filtering for this process instance. By analyzing the most probable process instances, the investigator reduces the amount of false positives he analyses. The link between the process instance and the associated red flags is only possible due to process mining. The fraud investigator can see the fraudulent process instance without switching to the ERP system to understand the as-is flow of this process instance.
- 2) Second, the fraud investigator can filter for fraud detection patterns and display the respective process instances where the specific fraud detection pattern occurs.
- 3) Third, the fraud investigator can analyze the event log and filter for process deviations. When filtering for a specific process instance, the red flags table on the GUI automatically displays only the red flags occurring in this process instance. If there are no red flags, the probability that fraud is in this specific instance reduces so there is no need for the investigator to analyze this process instance in detail (which reduces the number of false positives).

In all three cases, supporting information about the involved supplier(s), employee(s) and purchased material(s) is shown automatically in the GUI.

3.4. Evaluation of fraud detection

Phua et al. (2010) analyzed methods potentially useful for evaluating the various fraud detection approaches. “Most fraud departments place monetary value on predications to maximize cost savings/profit according to their policy” (Phua et al., 2010). As we aim to reduce the false positive rate, the monetary calculation is not suitable for our evaluation.

Further fraud detection evaluations include the use of heuristics (Hopwood et al., 2011). Another method to evaluate fraud is the ROC curve, which analyses the proportion of true positive and false positive values (Witten and Frank, 2005). The confusion matrix is often used for assessing the quality of the fraud detection classification. Although originating from data mining literature, this method is also used for fraud detection (e.g. Phua et al., 2010 or Hopwood et al., 2011). The confusion matrix is a specific table layout allowing a visualization of the performance of the fraud detection approach by showing true positive rate, false positive rate, true negative rate and false negative rate. We applied the confusion matrix as it shows the false positive rate that we aim to reduce.

3.5. Data generation

A large volume of realistic data with both fraudulent and non-fraudulent cases is necessary to detect fraudulent activities. Due to security concerns, very few companies are willing to provide their data to scientists, especially when fraud is suspected in the data (Yannikos et al., 2011). Hence, many researchers try to generate this data synthetically (e.g. Lundin et al., 2002; Barse et al., 2003 or Yannikos et al., 2011).

We created a proxy to collect semi-realistic fraud data by designing and conducting a serious game called the White Collar Hacking Contest (WCHC) (Schermann and Boss, 2014). In this contest, participants compete against each other as they, in turn, act as fraudsters and fraud examiners in an SAP ERP system. The fraudsters first develop fraud schemes and realize them in an SAP ERP

system. After the realization of the case, we swap the roles of the participants and asked them to analyze the business transactions and uncover fraud cases from the other teams. The contest was developed in close cooperation with fraud forensic experts from industry to ensure real-life conditions and fraud cases.

To win the contest, the students had to (1) generate fraud schemes that the other teams were not able to uncover, and (2) develop analytical capabilities to uncover fraud from other teams. To help generate interesting and close to real-life fraud schemes, each team first discussed its fraud ideas with a professional fraud investigator from industry (acting as a mentor) during the contest. The experienced fraud investigators discussed the students' ideas and presented interesting fraud cases from their professional career to help their team win the competition. The discussion helped to ensure that only realistic fraud cases were realized in the SAP ERP system.

We chose an SAP ERP system as the “playground” for the WCHC as this system is one of the worldwide market leaders for ERP systems (Pang et al., 2013). To ensure the generalizability of our approach, we used SAP ERP systems with different datasets and preconfigured customizing settings – IDES and GBI. IDES is an SAP virtual demo company with customizing settings and exemplary data for teaching and showcasing purposes. Global Bike Inc. (GBI) was first introduced by Magal and Word (2011) to teach integrated business processes with ERP systems. The SAP ERP system with the GBI dataset included comparably less predefined transactional data. The implemented fraud cases were of a general nature and could occur in any company, even if a different ERP system was used. Within an ERP system containing only fraud, the fraudulent process instances would be easy to detect. Therefore, we conducted the “normal” purchase-to-pay business process with a data generation tool (Baader et al., 2016).

Our approach is subject to some limitations. First, the authorization concept of both systems (IDES and GBI) is not realistic. Students were assigned the SAP-ALL profile, which allows execution of the whole purchase-to-pay business process with one user. Controls, like the Segregation of Duty Principles, were not implemented in the standard settings of either system. Second, the period of the execution of each process instance was very short: within only a few minutes or hours the whole purchase-to-pay business process was executed.

In general, data generation approaches use either the extrapolation of historic user data or the model of user behavior based on expert knowledge. An advantage of the extrapolation of historic user data (Lundin et al., 2002; Barse et al., 2003) is the authenticity of the dataset. However, the quality of the dataset is dependent on the sample dataset quality. We needed a dataset that included many different fraud cases. Therefore, we used the expert knowledge user simulation applied by e.g. Yannikos et al. (2011). Instead of only interviewing experts about the critical topic of fraud, we used the WCHC as a proxy to collect fraudulent process instances. The main advantage of our data generation approach is that we leave room for the students to be creative in terms of the fraud cases and we indirectly collect interesting fraud cases from practice.

4. Results

The analyzed dataset included 216,031 process instances of the IDES dataset and 161,101 process instances of the GBI dataset. Within the IDES dataset, the prototype identified 8 of 13 implemented fraud cases, whereas in the GBI dataset the prototype identified 7 of 18 implemented fraud cases. A detailed overview of all fraud cases implemented by the participants of the WCHC is shown in Table 4 (IDES) and Table 5 (GBI). In total, 15 of 31 fraud cases were identified successfully (true positive, TP). Although available, 16 fraud cases were not identified in the IDES and GBI dataset (false negative, FN) and 1399 cases were identified that were not implemented by the participants of the WCHC (false positive, FP). There were 377,101 process instances that were correctly analyzed as compliant (true negative, TN). Based on these results, we calculated a confusion matrix as described in Section 3:

$$TP = \frac{TP}{TP + FN} = \frac{15}{15 + 16} = 0.4838 \quad (1)$$

$$FN = \frac{FN}{TP + FN} = \frac{16}{15 + 16} = 0.5161 \quad (2)$$

$$FP = \frac{FP}{TN + FP} = \frac{1399}{377,101 + 12} = 0.00370 \quad (3)$$

$$TN = \frac{TN}{TN + FP} = \frac{377,101}{377,101 + 12} = 0.9999 \quad (4)$$

The resulting fraud confusion matrix is shown in Table 3.

We provide information about the fraud cases implemented by the teams and whether our prototype was able to identify the fraud

Table 3

Fraud confusion matrix.

	Fraud included in dataset	Fraud not included in dataset
Fraud identified by prototype	True positive: 48.38%	False positive: 0.37%
Fraud not identified by prototype	False negative: 51.61%	True negative: 99.99%

Table 4
Identified fraud cases in the SAP IDES system.

#	Fraud from WCHC	Fraud description	Identified?
1	IBAN fraud (redirect payment fraud)	Due to recent European regulations, bank and account numbers of all bank accounts were switched to IBAN and BIC numbers. After the switch was done correctly in the IDES company, an old bank code was changed to the one from an accomplice of the fraudster. The invoice was paid to the accomplice using the old bank account number.	Yes
2	Service fraud (shell company)	A cleaning service was ordered but not delivered by a shell company.	Yes
3	Returned goods fraud (shell company)	Gaskets were bought then returned to an accomplice vendor. The invoices were paid although the goods were sent back to the vendor.	Yes
4	Broken goods fraud (non-accomplice vendor)	Goods were delivered but wrongly claimed to be broken by the forklift driver at the company. The goods were sold by the employees on an internet platform.	No
5	Bidding fraud (bid rigging)	Computer chips were to be purchased. During a bidding process, the most expensive offer from the accomplice vendor was chosen although all submitted computer chips had the same quality.	Yes
6	Expensive motors fraud (pass-through fraud, shell company)	A motor was bought from a shell company. The motor was far more expensive than the average market price. The accomplice bought the motor for a normal price and resold it to his own company for a higher price.	Yes
7	Poor quality fraud (kickback fraud)	Goods with poor quality were delivered by an accomplice vendor but sold for the price of high quality goods. The fraudster responsible for quality assessment declared the goods as being of high quality. The overpaid amount of money was shared between the fraudster and the accomplice.	No
8	Shipping cost fraud (kickback fraud)	Copper was ordered from Jeddah (Saudi Arabia) for delivery in Hamburg (Germany). An extra charge was added for the transportation of the goods as the shipment route crossed an area known for sea piracy. However, the actual route taken was a safe route. The extra charge was shared between the fraudster and the accomplice.	No
9	Changing bank account (redirect payment fraud)	Before an invoice was paid, the bank account of the supplier had been changed to the one of the accomplice. The invoice was paid to the accomplice.	Yes
10	Overpriced tables (kickback fraud)	Overpriced tables were ordered from an accomplice vendor. The difference between the real price and the paid price was shared between the fraudster and accomplice.	Yes
11	No goods receipt fraud (shell company)	Goods were ordered but never received. Nevertheless, the invoice was paid.	Yes
12	Transportation cost fraud (kickback fraud)	Automobile taillights were bought from a supplier. The transportation costs per unit were slightly increased by the fraudster and paid by the recipient.	No
13	R&D extra charge (kickback fraud)	A special order of sensors with new requirements was sent to the supplier. An extra charge for development of the sensors was negotiated with the supplier for a certain period. After the end of this period, the extra R&D charge continued to be paid and shared between the fraudster and accomplice.	No

case (Table 4).

Our false negative rate is comparatively high. We were not able to identify five implemented fraud cases with our fraud detection prototype in the IDES dataset. One fraud case included increased shipping costs. Our literature review did not produce any fraud cases dealing with shipping costs. We will therefore add additional fraud detection patterns to our prototype to be able to identify shipping cost fraud. Our prototype was also unable to identify inferior quality items sold for a normal price.

Within the GBI system, we were able to identify 7 of 18 fraud cases. Kickback fraud is especially difficult to identify as there may be legitimate reasons for paying a price above the average or usual price. Possible reasons are better quality of the goods, poor negotiation skills of employees and an increase in the price of raw materials. Also wrong declamations are difficult to spot. This also increased our false negative rate.

5. Concluding discussion

The goal of our research was to reduce the number of false positives in internal fraud detection. The confusion matrix proved very useful for our research as it shows the false positive rate. However, in a real-world dataset, a complete confusion matrix cannot be calculated as the number of fraud cases that actually exist in the dataset is unknown. But true positive and false positive rates can be determined for both synthetic and real datasets.

We compare our results to some existing studies that report false positive and true positive rates. In the field of billing fraud, [Barse et al. \(2003\)](#) used neural networks to identify fraud. [Bezerra and Wainer \(2008\)](#) analyzed process aware information systems regarding fraud with an anomaly detection algorithm. Their results are summarized in Table 6.

Compared with the results of [Barse et al. \(2003\)](#), our approach performs better with both the true positive and false negative rates. The approach proposed by [Bezerra and Wainer \(2008\)](#) shows a very high true positive rate. Our approach, by contrast, shows a poorer fraud detection rate. However, our false positive rate is much lower than their rate. By adding additional fraud detection patterns we will be able to further increase our hit rate. Approaches with a false positive rate of over 50% can make the analysis overwhelming for

Table 5
Identified fraud cases in SAP GBI system.

#	Fraud from WCHC	Fraud description	Identified?
1	Bidding fraud (bid rigging)	Overprized products purchased by the vendor due to “better quality”, “better customer service” and “reliability of the purchaser”.	Yes
2	Go-green initiative	The company supported climate protection with a go-green initiative. Employees were offered bonuses for reduced CO2 output. A new truck was bought which had a manipulated CO2 emission. The employee received the bonus and shared it with the truck supplier.	No
3	Pre-payment fraud (shell company)	Pedals for new bicycles were ordered and prepaid. The delivering company declared insolvency; the total amount of the payment was not received by the GBI company.	Yes
4	Poor payment conditions	The new supplier offered advantageous payment conditions if payment was made early. The fraudster convinced his boss, who was unaware of the new conditions, to give him a bonus if he could renegotiate payment conditions.	No
5	Missing goods fraud (shell company)	Only a portion of the 20,000 purchased laptops was delivered but payment was issued for all laptops.	Yes
6	Poor quality fraud (bribe)	There were quality issues with ordered protective gear. The quality manager received a bribe for not thoroughly checking the quality of the goods.	No
7	Bidding fraud (bid rigging)	A bribe was received to choose a more expensive bidder in a bidding process for a production robot.	Yes
8	Greek E fraud (kickback fraud)	The Greek letter “E” looks identical to the Roman or Latin capital letter “E”. Two master data sheets for the material “WHEELS” were created, one using the Greek E and the other using the Roman E with different price tags. The more expensive Greek E products were selected for an order.	No
9	Overprized goods (kickback fraud)	The price for TVs was changed and then the TV was purchased for an overpriced amount.	Yes
10	Poor quality fraud (shell company)	Low quality bike frames were purchased. The labels on the frames had been changed to that of a manufacturer of higher quality frames.	No
11	Double shipping cost fraud (double payment fraud)	A new statue was ordered and the shipping costs were included in the price. Nevertheless, shipping costs were charged separately.	No
12	Non-received discount (kickback fraud)	A discount of 10% was given to a company, but the amount paid was without the discount. The amount of the discount was shared between the fraudsters and the accomplice vendor company.	Yes
13	Transportation cost fraud (kickback fraud)	Overpriced shipping costs. Instead of paying for 8000 km, the fraudster set the distance at 9000 km.	No
14	Kickback fraud (kickback fraud)	Raw material was purchased for a higher than usual amount. The difference in price was shared between the fraudsters at the vendor and supplier companies.	No
15	Payment without goods receipt (shell company)	Raw materials were purchased and the invoice paid without receiving the goods.	Yes
16	Food fraud (personal purchases)	Due to a health initiative, fruit baskets were bought for the company. The purchasing department bought large amounts of fruit baskets and privately resold some of them (e.g. through eBay).	No
17	Inflated working hours (payroll fraud)	A new storehouse had to be built. The accomplice of the fraudster helped with site clearing. The number of hours actually worked was less than the number invoiced.	No
18	No early payment discount (kickback fraud)	Fraudster delayed the payment of a certain good to prevent his company from receiving an early payment discount. The early payment discount was still provided by the accomplice and the difference shared between the fraudster and accomplice.	No

Table 6
Comparison of research results on identifying fraud.

		True positive	False positive
Barse et al. (2003)	Synthetic dataset	26.6%	0.8%
	Real dataset	4.9%	6.5%
Bezerra and Wainer (2008)		99.9%	57.4%
Our fraud results		48.38%	0.37%

an analyst because a flood of alarms can be expected (Chandola et al., 2009).

Our false positive rate is very low compared to other research studies mentioned here. One reason for the very low false positive rate may be the underlying dataset. We assume that the dataset might be “cleaner” than in a real company as the participants of the WCHC followed the guidelines of the IDES and GBI curricula to execute the business process. Applying the prototype to a real dataset should help to validate the results.

Compared to the identification of red flags standalone, our approach shows several advantages. First, the fraud investigator is able to analyze the fraudulent as-is process instance on the dashboard without having to switch to the ERP system. Second, the link between process instances and red flags enables filtering for deviating process instances and displaying the occurring red flags and

vice versa (e.g. display process instances with a high number of red flags). Third, the search for red flags in combination reduces the high number of false positives.

Our research is subject to particular limitations. First, we applied a deductive approach and only previously known fraud detection patterns were identified in the dataset. Respectively, we aim to add further detection patterns to be able to identify more cases of fraud in the dataset. The architecture of this tool is designed to be easily extendable. Our vision is to enable any fraud investigator to add fraud detection patterns to steadily increase the database.

Second, our dataset contains close to reality fraud detection patterns. However, to prove validity using a real dataset, the approach should be applied to a dataset from a company.

We contribute to existing fraud detection literature by demonstrating the usefulness of combining flag- and pattern-based approaches. We also contribute through our grouping of different red flags to fraud detection patterns.

Our prototypical implementation may be useful for practitioners. For example, they may apply our prototype to their dataset to reduce the flood of alerts, which inherently includes many false positives. The approach discussed should be further validated with a real dataset from a company. It may also be beneficial to include a larger number of business processes, such as the sales process. Adding further fraud detection patterns to the prototype may increase our true positive rate.

Appendix 1. Corresponding red flags based on the literature survey results

This appendix contains the identified fraud detection patterns with the corresponding red flags.

Fraud type	Kickback fraud
Description	This scheme involves the collusion of vendors and employees, preferably authorized to approve invoices (Wells, 2011a, 2011b). While the vendor issues and submits invoices that are usually either inflated or fictitious, the insider makes sure the invoices are approved for payment and a share of the generated surplus is diverted back to? (Wells, 2011a, 2011b).
Red flags	Supplier 1. Sudden business activity with old “sleeping” supplier (sudden activity in non-active accounts) 2. High purchasing volumes from new supplier 3. Sudden activity in non- active accounts 4. Slow delivery 5. Supplier provides no typical discounts or special offers 6. Supplier is also customer 7. Only small number of suppliers 8. A supplier only invoices services Purchase requisition and purchases 9. Higher purchasing volumes seemingly not associated with higher business activity or an increase in stock (rising expenditure on goods and services) 10. Several small orders of the same product (a purchase is divided into several smaller purchases in order to bypass the approval process) 11. Unclear reason for ordering or few details about the goods received 12. Purchasing value significantly exceeds the last value 13. Sudden activity in non-active accounts 14. No approval for order Goods received 15. No goods receipt document 16. Unusual high stocks and increased purchases from a specific supplier Invoice and payment 17. Overpayment of purchased products or services (purchases are above market price) 18. Frequent payments of the same (rounded) amount to a supplier 19. Money transactions outside normal business hours 20. Invoice documented outside normal working hours 21. Double payments 22. Supplier is regularly paid faster than other suppliers (invoices are settled very quickly) 23. Payments exceed the average for a vendor 24. Payments exceed the total average 25. The order amount is higher than the invoice 26. Supplier invoice is higher than the order amount

Sources	Allen, 2007; Brandman, 2000; Brulenski and Zayas, 2004; Buckhoff and Parham, 2009; Cowan, 2005; Ellinor, 2009; Fishman, 2001; Graycar and Sidebottom, 2012; Grieshober, 2001; Johnson and Rudolph, 2009; Kranacher, 2008; Morehead, 2007; Sammons, 2005; Viton, 2003; Wells, 2002c; Wells, 2003a
Fraud type	Bid rigging
Description	Occurs when vendors are willing to pay for influencing a competitive bidding process (Wells, 2011a, 2011b) and ranges from swaying the specification in the pre-solicitation phase to making competitors' bids accessible before the submission phase ends (ACFE, 2016).
Red flags	Supplier 1. The same person approves the new supplier as well as payments to the supplier 2. Fictional bidder 3. Supplier often receives excessive surcharge 4. The same supplier bids under different names Call for tender 5. The winner of the tender process is always the last bid of a tender 6. Companies have the opportunity to change their offers 7. Strong limitation of the time period for bidding 8. The offer is very specific/The number of bidders is relatively small (possible limitation of competition) 9. Several small calls for tenders (a tender is divided into several smaller calls for tenders in order to bypass the approval process) 10. Tender offers are very similar 11. Huge outlier in the bids (price, quality etc.) 12. When a new supplier joins the auction, the bid prices start to fall 13. Offers accepted after the end of the bidding phase 14. Wording of contract favors one supplier 15. Missing transparency 16. Supplier is always chosen without a clear competitive advantage 17. Qualified supplier does not provide bids (no invitation to the tender process to limit competition) 18. Same supplier bid for every project or product bid (suspicious if there are different products) 19. Publishing of tenders in insignificant publications or during holiday times to limit competition 20. Favorable bid but with numerous change requirements (possible collusion with supplier) Purchase requisition and purchase 21. Goods are purchased without comparing competitive offers 22. Order amounts are just below authorization thresholds: no approval required 23. Changes to cost specifications after order placement Invoice and payment 24. Invoices for never delivered goods 25. Prices for purchased goods or services are above current market prices 26. Unjustified pre-payments
Sources	Brandman, 2000; Cowan, 2005; Ellinor, 2009; Fishman, 2001; Graycar and Sidebottom, 2012; Grieshober, 2001; Huntington et al., 1999; Johnson and Rudolph, 2009; KPMG, 2010; Lambert-Mogiliansky and Sonin, 2006; Lander et al., 2008; May, 2005; McNeal, 2012; Morehead, 2007; Pacini and Brody, 2005; PWC, 2016; Sammons, 2005; Thai, 2001; Viton, 2003; Wells, 2003a, 2013; Zikmund, 2008
Fraud type	Shell company
Description	Shell company is a fictive entity without active business activities or significant assets. This is not necessarily illegal, but it is assumed that the shell company was founded solely for committing an economic criminal offense. In order to be able to receive payments, a bank account is usually set up on behalf of the shell company.

Red flags	Supplier
	1. Sudden business activity with old “sleeping” supplier (sudden activity in non-active accounts) 2. High purchasing volumes from new supplier 3. One-time supplier 4. The same person records a new supplier within the system and orders from this supplier 5. Missing data in supplier's master data record (e.g. no active telephone number) 6. P.O. box as only address of supplier 7. Supplier invoices only services 8. Name of the supplier consists exclusively of initials 9. Initial letters of an employee's name correspond to a supplier's name 10. Supplier master data corresponds to data of an employee (address, bank data, etc.) 11. One-time supplier 12. Several records of the master data in the supplier list or database (supplier with same name, telephone number or address)
	Purchase requisition and purchase order
	13. No authorization for order 14. Unusual authorization (e.g. above average number of orders per day approved) 15. Sales employees place “urgent” orders 16. Purchase orders are made before the purchase applications are approved 17. Unclear ordering reasons or few details about the service 18. Order amounts are just below authorization thresholds: no approval required 19. Purchases are divided into several partial purchases in order to bypass the approval process 20. Purchasing value exceeds the last value by a significant amount 21. Unusual purchasing amount from one supplier
	Goods receipt
	22. Invoice for undelivered goods/services 23. No examination of goods receipt through an independent employee
	Invoices and payment
	24. Excessive invoices from one supplier/increasing amount of invoices 25. Tax missing on invoice 26. Sequential invoice numbers from one supplier 27. Multiple invoices from one supplier per month, although a monthly payment would be usual 28. No or incorrect employee ID on the invoice 29. Invoices from one particular supplier always approved by the same employee 30. Invoices with the same (usually rounded) amount 31. Actions performed at unusual times (e.g. outside normal business hours) 32. Invoices settled very quickly 33. Overpayment of related products or services 34. Supplier invoice is higher than order amount 35. Order amount is higher than invoice
Sources	Barron, 2011 ; Brandman, 2000 ; Buckhoff, 2002, 2003 ; Christensen and Byington, 2003b ; Ellinor, 2009 ; Fishman, 2001 ; Johnson and Rudolph, 2009 ; Langley, 2003 ; Lehman, 2008 ; Lehman and Weidenmier, 2005 ; May, 2005 ; Meiners, 2005 ; Nilsen, 2010 ; Sammons, 2005 ; Viton, 2003 ; Wells, 2002a ; Wells, 2003b, 2004 ; Zikmund, 2008
Fraud type	Double payment
Description	An attempt is made to pay out the invoice several times. The payment is often made twice to an accomplice of the fraudster. An already paid invoice can be repaid.

Red flags	Supplier
	1. P.O. box as only address of supplier 2. Supplier master data corresponds to data of an employee (address, bank data, etc.)
	Purchase requisition and purchase
	3. Duplicate information (same purchase to different supplier)
	Invoice and payment
	4. Multiple different invoices for same goods 5. Exactly the same invoice amount is paid to two different suppliers 6. The same invoice number on two different documents 7. Same-same-same test (same person pays the same supplier on the same day the same amount) 8. Same-same-different test (different person pays the same supplier on the same day the same amount) 9. Same invoice number, same supplier, different amount 10. Anomalies in invoice amounts (fictitious numbers that do not correspond to any mathematical laws (e.g. Benford's Law))
Sources	Christensen and Byington, 2003a ; Islam et al., 2010 ; Langley, 2003 ; Pacini and Brody, 2005 ; Taylor, 2006
Type of fraud	Pass through
Description	Perpetrators sell goods or services at inflated prices to their own company. The perpetrator buys the product or service and resells it at a higher price.
Red flags	Supplier
	1. High purchasing volumes at a new or unauthorized supplier
	Purchase requisition and purchase
	2. Higher purchasing volumes that cannot be explained with higher business activities or a higher stock (increasing expenses for goods and services)
	Goods receipt
	3. Unusual high inventory combined with the corresponding purchases at a certain supplier (unnecessarily high/increasing inventories)
	Invoice and payment
	4. Extensive budget deviations 5. Overpayment of products or services (purchase prices are above market prices) 6. The same invoice number on two different documents 7. Same-same-same test (same person pays the same supplier on the same day the same amount) 8. Same-same-different test (different person pays the same supplier on the same day the same amount)
Sources	Christensen and Byington, 2003b ; Nilsen, 2010 ; Viton, 2003 ; Wells, 2002b ; Wells, 2003b, 2003c, 2004
Type of fraud	Non-accomplice vendor
Description	A legitimate supplier not involved in the fraud case is used to defraud the company. For example, an invoice from a legitimate supplier is overpaid by the perpetrator. Then the additional amount is requested to be returned (e.g. on the pretext of an accounting error). The repayment is intercepted by the perpetrator before it reaches the receiving company. Another method is to deliberately declare a wrong supplier as a payee in order to steal the withheld money. A further form of this scheme involves the purchase of unused goods, their subsequent return and the interception of the credit note issued by the supplier for the returned goods. In rare cases, even the claims of legal suppliers are paid directly to the perpetrator's shell company.

Red flags	Supplier
	1. Suspicious address of the supplier: same address as another supplier 2. High purchasing volumes for a new or unauthorized supplier Goods receipt 3. Return of goods Invoice and payment 4. Supplier invoice is higher than the order amount 5. Order amount is higher than invoice 6. Overpayment of purchased products or services 7. The exact same purchase value is paid to two different suppliers 8. Money transactions at unusual times (outside normal business hours) 9. Multiple different invoices for the same goods
Source	Christensen and Byington, 2003b ; Nilsen, 2010 ; Viton, 2003 ; Wells, 2003b, 2004
Fraud type	Redirect payment fraud
Description	Legitimate transactions are manipulated in such a way that the payment is transferred to the perpetrators bank account. The employee or an accomplice must have the authorization to change the master data of the vendor. After changing the master data (e.g. bank account), the perpetrator attempts to cover his tracks by changing the bank account information back to the original information.
Red flags	Supplier 1. Changes in the master data (e.g. bank account) (mostly before payment) 2. Determination of the supplier for whom the invoice verification is deactivated 3. Double information Invoice and payment 4. Large budget deviations 5. Supplier invoice is higher than order amount 6. Adjustments to liabilities (in accounts payable) 7. A payment recipient with a name similar to the previous name is entered 8. Change of currency between purchase and payment to take advantage of conversion differences 9. Doubling and redirecting the invoice 10. Same invoice number, same supplier, different amount
Source	Islam et al., 2010
Type of fraud	Personal purchases
Description	Making private purchases at the expense of the company. Most perpetrators buy goods or services for their personal use and then erroneously record the invoice as a liability in the company's internal accounting. In order to conceal the actual purpose of the purchase, purchased goods are declared as business requirements or fake invoices are submitted. As in the case of a shell company, the perpetrators are often responsible for authorizing purchase orders. The perpetrator may either keep the purchases or return them and keep the money. A different form of this schema does not require an authorization of the payment: private purchases with the company-owned credit or purchasing card.

Red Flags	Purchase requisition and purchase
	1. Order amounts are just below authorization thresholds: no approval required 2. Multiple small purchases of the same product 3. Imprecise reason for the received good or service
	Invoice and payment
	4. Invoices, receipt confirmation and order documents do not match 5. Overpayment of purchased products or services (purchases are above the market price) 6. No or incorrect employee ID on the invoice 7. Invoice receipt although the purchase request has been blocked 8. Invoice receipt although the purchase request has not been authorized 9. Unusual authorizations 10. Same-same-same test (same person pays the same supplier on the same day the same amount) 11. Same-same-different test (different person pays the same supplier on the same day the same amount)
Source	Christensen and Byington, 2003b; Nilsen, 2010; Viton, 2003; Wells, 2003b, 2004

Appendix 2. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.accinf.2018.03.004>.

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